

Sneha Hassan

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SKILLS

AI & ML: LLM Fine-tuning, Transfer Learning, PyTorch, TensorFlow, LangChain, RAG, Agentic Workflows, NLP, OpenCV, Signal Processing, Model Evaluation, Inference Optimization

Languages: Python, TypeScript, JavaScript, Java (Spring Boot), SQL, C

Backend & APIs: RESTful APIs, Flask, Node.js, Microservices, Event-Driven Architecture, GraphQL, Firebase

Cloud & DevOps: AWS, GCP, Docker, CI/CD, Git, Linux, Firebase Cloud Functions

Data: PostgreSQL, MongoDB, Redis, Data Pipelines, Schema Design, Query Optimization

Frontend: React.js, Next.js, React Native, TypeScript, HTML/CSS

EXPERIENCE

Carnegie Mellon University

Feb 2026 – Present

Graduate Researcher, GenAI & Bio-Imaging

Pittsburgh, PA

- Architected and deployed a custom 8B-parameter LLM with tool-calling capabilities for Cryo-Electron Tomography, automating cellular annotation workflows that previously required 8 hours of expert manual labeling per dataset
- Built async model evaluation infrastructure and automated grading system across 296 bioinformatics benchmarks, enabling systematic regression testing of fine-tuned model variants
- Optimized LLM inference pipelines using Python batch processing and caching, achieving a 3.7x throughput improvement (45s to 12s per sample) while maintaining output quality

Ather Energy Ltd. (EV Scooter Company)

Oct 2023 – Nov 2024

Software Developer

Bangalore, India

- Trained and deployed a production ML pothole detection model (92% accuracy) using Python signal processing on raw accelerometer sensor data, owning the full pipeline from data cleaning to model deployment on a 50K+ DAU system; drove 35% safety-feature adoption growth
- Engineered and shipped a full-stack PWA using Spring Boot and React Native for carbon-footprint tracking, serving 50K+ DAU and reducing end-to-end latency by 25% through API response caching
- Architected a B2B fleet monitoring dashboard with real-time IoT telemetry (GPS, remote immobilizer, diagnostics) using React and Firebase, adopted by 3 enterprise clients managing 200+ vehicles
- Built cross-team test automation platform in Java, increasing automated test coverage from 32% to 78% and cutting QA cycle time by 45% across 4 engineering squads

Ather Energy Ltd.

Feb 2023 – Oct 2023

Software Developer Intern

Bangalore, India

- Developed 6 RESTful APIs in Spring Boot powering a favorite-locations feature for the Ather app (100K+ users), cutting location-search latency by 40% with indexed queries and Redis caching; built a full-stack campaign PWA driving 25K+ sign-ups in the first month

Telvue Corporation (Broadcast Automation)

May 2025 – Aug 2025

Software Developer Intern

Mt. Laurel, NJ

- Designed and shipped an event-driven push notification system using Rails microservices and Firebase Admin SDK with pub/sub triggers and retry logic, increasing mobile user engagement by 40% across 13K+ users; led 4-person cross-functional team reducing integration cycle from 3 weeks to under 2 weeks

PROJECTS

Socratic LLM for Technical Pedagogy

Oct 2025 – Feb 2026

- Fine-tuned a foundation LLM with custom NLP guardrails to deliver Socratic hints instead of answers; applied NLP quality metrics to eliminate answer-dumping by 85% in a 30-user pilot; built full-stack app (React and Flask) with under 2s inference latency

Vending Analytics | Software Engineer, Sheetz

Aug 2025 – Dec 2025

- Engineered a computer vision pipeline (Python, OpenCV, TensorFlow) detecting empty vending slots at 94% accuracy, served via a React dashboard that cut stockout response time by 60%

RecoVR – VR Rehabilitation Platform | Founder & Software Engineer, Incubated by CMU

Oct 2023 – Present

- Built a Unity VR physiotherapy app with custom C# scripts and real-time positional tracking, improving patient fitness outcomes by 40% vs. traditional home exercises in a 20-patient pilot study

EDUCATION

Carnegie Mellon University

Jan 2025 – Dec 2025

M.S. in Engineering & Technology Innovation Management | Software and Data Focus | GPA: 3.77/4

Pittsburgh, PA

APJ Abdul Kalam Technological University

Aug 2019 – May 2023

B.Tech in Computer Science | GPA: 3.94/4

Kerala, India

LEADERSHIP & AWARDS

United Nations Millennium Fellow — Led 150 global fellows for community-impact work on SDG 8 (2022–2023)

Teaching Assistant — Product Management for Innovation, Carnegie Mellon School of Computer Science